

SIMATS ENGINEERING

CO5-ASSESSMENT TOOL3-Reverse Engineering

COURSE NAME: Artificial Intelligence

COURSE CODE: CSA1709

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COURSE OUTCOME COVERED

CO5: Design intelligent agents and real-time AI-based systems (chatbots, expert systems, emotion detection, edge AI, autonomous drones, and related applications) by selecting appropriate agent architectures, knowledge representation, and reasoning mechanisms to solve real-world problems. (BL6)

Assessment Tool Weightage: 20%

QUESTIONS: 5

Total Marks:50

Q1. The following is a conversation between a user and an AI-based flight-booking chatbot. Study the transcript given below and reverse-engineer the chatbot's design by: (a) writing its PEAS description, (b) classifying the type of agent it represents with justification, (c) identifying the likely component (NLU, Dialogue Manager, Knowledge Base, or NLG) responsible for each chatbot response, and (d) suggesting one enhancement using a learning-agent architecture.

Speaker	Message
User	I want to book a flight to Chennai next Friday.
Chatbot	Sure! What time would you like to depart, and do you have a preferred airline?
User	Morning flight, no preference.
Chatbot	I found 3 morning flights to Chennai on Friday. Shall I show you the cheapest option first?

Q2. The block diagram below shows the processing pipeline of a Real-Time Emotion Detection System. Reverse-engineer the system by: (a) explaining the function performed at each stage of the pipeline, (b) identifying the likely AI/ML technique used at each stage, (c) writing the PEAS description of the system, and (d) classifying the

Real-Time Emotion Detection System — processing pipeline



agent type (e.g., simple reflex, model-based) with justification.

Q3. An Expert System Shell for medical diagnosis contains the rule base, input facts, and reasoning trace given below. Study it and reverse-engineer the shell by: (a) identifying the type of chaining (forward/backward) used to reach the diagnosis, with justification, (b) identifying the expert-system-shell components illustrated (knowledge base, inference engine, explanation facility), (c) explaining how the shell would perform knowledge acquisition to add a new rule for dengue, and (d) explaining, with reasoning, whether backward chaining could reach the same diagnosis.

Rule	IF (condition)	THEN (conclusion)
R1	fever = yes AND cough = yes	suspect = flu
R2	suspect = flu AND body_ache = yes	diagnosis = Influenza
R3	fever = yes AND rash = yes	suspect = measles

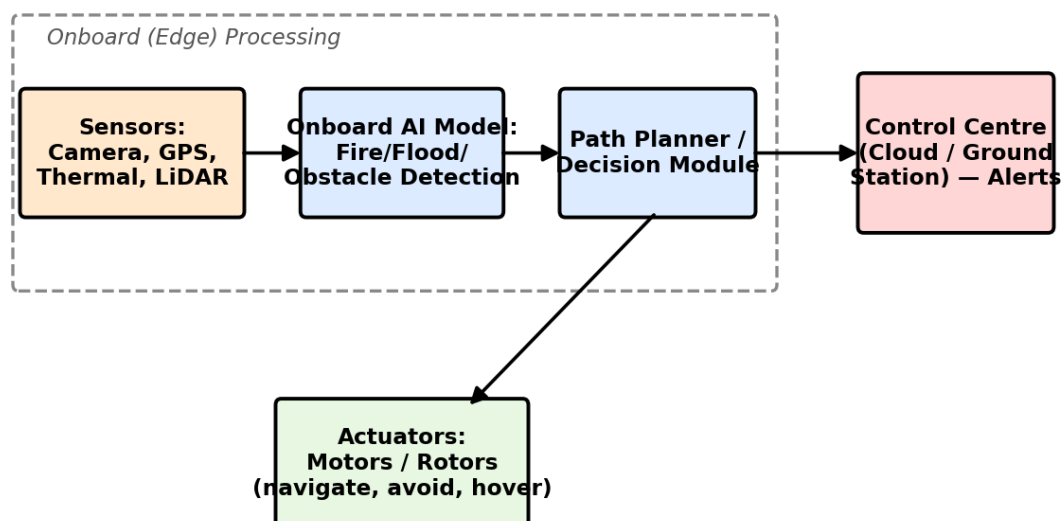
Given facts: fever = yes, cough = yes, body_ache = yes

Reasoning trace: R1 fires first (fever & cough matched) → suspect = flu. R2 then fires (suspect = flu & body_ache matched) → diagnosis = Influenza.

Explanation generated by the shell: “Diagnosis = Influenza because fever and cough indicated flu, and flu together with body ache confirmed Influenza.”

Q4. The diagram below shows the system architecture of an Autonomous Drone used for Disaster Detection with Edge AI deployment. Reverse-engineer the design by: (a) writing the PEAS description of the drone agent, (b) classifying the agent type (goal-based/utility-based) with justification, (c) identifying which components run on the edge versus the cloud and justifying why on-edge processing was chosen for detection and obstacle avoidance, and (d) identifying whether the agent's control architecture is reactive, deliberative, or hybrid, with justification.

Autonomous Drone — Disaster Detection (Edge AI) system architecture



Q5. The table below shows sample output logs from an AI-based Deepfake & Face Authenticity Verification system across five video frames. Study the output and reverse-engineer the system by: (a) identifying the likely input features used by the model (e.g., blink rate, texture/artifact score), (b) identifying the type of model/technique likely used to produce these outputs, (c) explaining how the final Real/Fake decision may have been derived from the frame-level confidence scores,

and (d) suggesting one way a swarm/ensemble-based approach could improve the system's reliability.

Frame No.	Blink Rate (per 10s)	Texture/Artifact Score	Fake Confidence (%)	Prediction
1	4	0.12	8	Real
2	1	0.61	74	Fake
3	0	0.83	91	Fake
4	5	0.09	5	Real
5	2	0.55	68	Fake

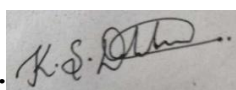
Overall system verdict: Fake (3 of 5 frames flagged — majority vote)

Code of Conduct:

I K.S.DHARSHANA(192412015)certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:



RUBRICS FOR CALCULATION

Criteria	Level 4 - Exemplary	Level 3 - Proficient	Level 2 - Developing	Level 1 - Beginning	Max Marks
Identification of Agent Type & PEAS (correctly inferred from the given artifact)	Correctly identifies the agent type and gives a complete, accurate PEAS description, fully consistent with the given artifact.	Identifies the agent type and PEAS description correctly with only minor omissions.	Partially correct identification of agent type/PEAS; some elements are missing or inconsistent with the artifact.	Agent type/PEAS identification is largely incorrect or not attempted.	10
Identification of Architecture & Components (mapping artifact evidence to system components)	Accurately identifies all relevant architectural components and correctly maps each to specific evidence in the artifact.	Identifies most components correctly with reasonable mapping to the artifact; minor gaps present.	Identifies some components; mapping to the artifact is weak or partially incorrect.	Little or no correct identification of architecture/components.	10
Identification of Underlying Technique/Algorithm	Correctly reverse-engineers the	Correctly identifies the	Identification is only partially	Technique/mechanism is incorrectly	10

Criteria	Level 4 - Exemplary	Level 3 - Proficient	Level 2 - Developing	Level 1 - Beginning	Max Marks
Algorithm/Reasoning Mechanism	technique, algorithm, or reasoning mechanism (e.g., chaining type, ML technique) with clear supporting evidence.	technique/mechanism with adequate, mostly accurate justification.	correct or lacks sufficient supporting evidence.	identified or not addressed.	
Quality of Justification & Use of Evidence (linking conclusions back to the given artifact)	Every conclusion is clearly justified with specific, accurate references to the given artifact.	Most conclusions are justified with reasonable references to the artifact.	Justification is vague or only loosely connected to the artifact.	Conclusions are stated with little or no justification or evidence.	10
Clarity & Completeness of Presentation (addresses all sub-parts of the question in an organised manner)	All sub-parts are addressed completely, in a clear, well-organised, and logically structured manner.	Most sub-parts are addressed clearly with minor gaps in organisation or completeness.	Some sub-parts are missing or the presentation is unclear/poorly organised.	Response is incomplete, disorganised, or largely missing.	10
				Total	50

Q1. Reverse Engineering of AI-Based Flight Booking Chatbot

(a) PEAS Description

P – Performance Measure:

The chatbot should understand the user's requirement correctly and help them find and book a suitable flight. It should give correct flight details, respond quickly and reduce mistakes during booking.

E – Environment:

The environment includes the user, flight schedules, airlines, available seats, timings and ticket prices.

A – Actuators:

The chatbot gives replies, asks questions, displays available flights and can help the user proceed with booking.

S – Sensors:

The main sensor is the user's message. From the message, it gets details such as destination, date, time and airline preference.

(b) Type of Agent

The chatbot can be considered a **goal-based agent**. Its main goal is to help the user complete a flight booking. It first understands that the user wants to travel to Chennai next Friday. Since the departure time and airline are not given, it asks for those details. After receiving "Morning flight, no preference", it searches for suitable flights.

So, the chatbot does not simply give a fixed reply. It uses the information provided by the user to move towards the goal of finding a suitable flight.

(c) Components Responsible for the Responses

- **NLU:** Understands the user's message and identifies details such as Chennai, next Friday and morning.
- **Dialogue Manager:** Decides what to ask next. This is why it asks about departure time and preferred airline.
- **Knowledge Base:** Contains or accesses flight information such as timings, airlines and prices.
- **NLG:** Converts the system's decision into a normal sentence that the user can understand.

For example, the reply "**I found 3 morning flights to Chennai on Friday**" needs flight information from the knowledge base, while the final sentence is generated through NLG.

(d) Learning-Agent Enhancement

A learning component can be added to the chatbot so that it learns from previous user choices. For example, if a user usually chooses morning flights or cheaper tickets, the chatbot can learn these preferences and give more suitable options in future.

It can also learn from whether users accept or reject its recommendations. This will make the chatbot more personalized and improve its performance over time.

Q2. Reverse Engineering of Real-Time Emotion Detection System

(a) Function of Each Stage

The system works as a pipeline where the input passes through different stages.

1. Input:

The system receives real-time information, mainly through a camera or other suitable sensor.

2. Pre-processing:

The captured input is cleaned and prepared for further processing. For an image, operations such as resizing and normalization can be performed.

3. Feature Extraction:

Important features related to facial expressions are identified. These may include the position of the eyes, eyebrows, mouth and other facial characteristics.

4. Emotion Detection:

The extracted features are given to an AI/ML model. The model predicts the emotion, such as happy, sad, angry, surprised or neutral.

5. Output:

The detected emotion is displayed as the final result.

(b) Likely AI/ML Techniques

A **CNN (Convolutional Neural Network)** is a suitable technique for facial emotion detection because it can learn useful visual features from facial images. The complete process can be represented as:

Camera Input → Pre-processing → Feature Extraction → CNN Model → Emotion Prediction

Pre-processing may use image resizing and normalization, while the CNN performs feature learning and classification.

(c) PEAS Description

Performance Measure:

High detection accuracy, quick response, fewer wrong predictions and stable real-time performance.

Environment:

The person, surroundings, lighting, background and camera position form the environment.

Actuators:

The system can display the detected emotion or send the result to another application.

Sensors:

A camera is the main sensor used to capture the person's facial expression.

(d) Agent Type

The system can be classified as a **model-based agent**. It receives information from the camera, processes it and uses a trained model to determine the person's emotional state.

It is not just giving a fixed response to every input. It analyses the current input using the learned model and produces the most suitable emotion prediction. Therefore, a model-based architecture is suitable for this real-time emotion detection system.

Q3. Reverse Engineering of Medical Expert System

(a) Type of Chaining

The system uses **forward chaining**.

Forward chaining starts with the facts that are already available and then checks which rules can be applied.

Given facts are:

- fever = yes
- cough = yes
- body_ache = yes

First, **R1** is applied:

fever = yes AND cough = yes → suspect = flu

So the new fact becomes:

suspect = flu

Then R2 is applied:

suspect = flu AND body_ache = yes → diagnosis = Influenza

Thus, the final diagnosis is **Influenza**. This matches the reasoning trace given in the question, where R1 fires first and R2 fires next.

(b) Expert System Shell Components

Knowledge Base:

It contains the medical rules such as R1, R2 and R3.

Inference Engine:

It checks the available facts against the rules and decides which rule should be applied. In this case, it first applies R1 and then R2.

Explanation Facility:

It explains how the final diagnosis was obtained. The given explanation connects fever and cough with flu and then connects flu and body ache with Influenza.

(c) Adding a New Rule for Dengue

To add a dengue rule, knowledge can first be collected from a medical expert.

The expert's knowledge is then converted into an IF–THEN rule.

For example:

IF fever = yes AND rash = yes AND body_ache = yes

THEN suspect = dengue

The steps would be:

1. Collect information from the medical expert.
2. Identify the important symptoms.
3. Convert them into an IF–THEN rule.
4. Check the rule with the expert.
5. Add the rule to the knowledge base.
6. Test the rule using suitable facts.

This allows the expert system to handle dengue-related cases also.

(d) Can Backward Chaining Reach the Same Diagnosis?

Yes, backward chaining can also reach the same diagnosis.

In backward chaining, the system starts with the goal:

diagnosis = Influenza

It then checks R2 and finds that two conditions are required:

- suspect = flu
- body_ache = yes

Body ache is already given as a fact. To prove suspect = flu, the system checks R1. R1 requires fever and cough, and both are already given.

Therefore, the system can prove:

fever + cough → flu → Influenza

So both forward and backward chaining can produce the same diagnosis, but forward chaining starts from the facts whereas backward chaining starts from the required conclusion.

Q4. Reverse Engineering of Autonomous Drone with Edge AI

(a) PEAS Description

P – Performance Measure:

The drone should detect disaster situations correctly, avoid obstacles, reach the required location safely and complete its mission with minimum delay and energy usage.

E – Environment:

The environment includes damaged buildings, people, trees, obstacles, weather conditions and other objects present in the disaster area.

A – Actuators:

The drone uses its motors and propellers to control its movement, direction, speed and altitude.

S – Sensors:

The drone can use cameras and other onboard sensors to detect objects and understand its surroundings.

(b) Agent Type

The drone can be considered a **utility-based agent**.

The drone does not only have to reach a particular destination. It must also consider safety, obstacle avoidance, detection accuracy, battery level and mission time.

For example, if there are two possible paths, the drone should prefer the safer path even if it takes slightly longer. Therefore, it has to balance different factors, which makes a utility-based agent suitable.

(c) Edge and Cloud Components

Edge side:

- Camera and sensor processing
- Disaster detection
- Object detection
- Obstacle avoidance
- Immediate flight-control decisions

Cloud side:

- Storage of collected data
- Detailed analysis
- Model training or updating
- Maintaining mission information

Detection and obstacle avoidance are better performed on the **edge** because they need a quick response. If the drone sends all the data to the cloud and waits for a response, network delay can cause problems, especially in a disaster area where communication may not be reliable.

By processing the information on the drone itself, it can react immediately to an obstacle.

(d) Control Architecture

The drone uses a **hybrid architecture**.

The **reactive part** handles immediate situations such as detecting an obstacle and changing direction.

The **deliberative part** is used for higher-level tasks such as route planning, selecting targets and deciding how to complete the mission.

Therefore, both reactive control and planning are needed. This combination makes the architecture hybrid.

Q5. Reverse Engineering of Deepfake and Face Authenticity Verification System

(a) Likely Input Features

From the given table, the system clearly uses features such as:

- Blink rate
- Texture/artifact score
- Facial appearance
- Possible facial movement and consistency between frames

The table shows that frames with higher artifact scores generally have higher fake confidence. For example, frame 3 has an artifact score of **0.83** and fake confidence of **91%**.

(b) Likely Model/Technique

The system is likely using a **machine-learning/deep-learning classification model**.

A CNN can be used to analyse facial images and detect texture or visual artifacts. Since the input is a video, the system can also analyse changes between frames to identify unnatural facial movements.

The model gives a **fake confidence percentage** for each frame. A higher percentage means that the model considers that frame more likely to be fake.

(c) Final Real/Fake Decision

The frame results are:

Frame	Fake Confidence	Prediction
1	8%	Real
2	74%	Fake
3	91%	Fake
4	5%	Real
5	68%	Fake

There are **3 Fake predictions** and **2 Real predictions**.

The system uses **majority voting**:

Fake = 3 frames

Real = 2 frames

Since the number of Fake predictions is higher, the final result is:

Overall Verdict: FAKE

This is also the verdict directly given in the question: **3 out of 5 frames were flagged as Fake.**

(d) Swarm/Ensemble-Based Improvement

The reliability can be improved by using multiple models instead of depending on only one model.

For example:

- One model can check facial texture.
- Another can check blinking and facial movement.
- Another can check changes between video frames.
- Another can check facial landmarks.

Each model gives its own result, and the system can combine the results using majority voting or weighted voting.

This makes the final decision more reliable because one model making a wrong prediction will have less effect on the final result. It can also help the system detect different types of deepfake manipulation more effectively.